Scientific African 19 (2023) e01485

Contents lists available at ScienceDirect

Scientific African

journal homepage: www.elsevier.com/locate/sciaf

Addressing food insecurity in Richmond area of KwaZulu-Natal, South Africa: The role of cash transfers

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ARTICLE INFO

Article history:
Received 14 July 2022
Revised 13 November 2022
Accepted 29 November 2022

Editor Name: DR B Gyampoh

Keywords:
Cash transfer
Household food (in)security
HFIAS
Endogenous switching poisson model
South Africa

ABSTRACT

Food insecurity is one of the national priorities in South Africa. Cash transfer programs are hypothesized to alleviate short-term deprivations, regularize consumption, and reduce the adoption of negative coping strategies. This study estimates the role play of cash transfers in enhancing food security in South Africa. The household food security access scale (HFIAS) was used to estimate food security status of 180 sampled households. An endogenous switching poisson model was used to examine how household food insecurity was affected by access to cash transfers. The findings indicate that the household food security was statistically significantly influenced by access to cash transfers, household head gender, access to finance, access to farm-based organizations (FBO), access to cooperatives, and agricultural training. As a result, this study recommends the South African government and other donor organizations to expand cash transfer programs among low-income households in rural South Africa. Access to cash transfers is a key component of achieving enhanced household food security.

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Introduction

According to estimates, more than 690 million people experienced hunger globally in 2019, and the COVID-19 pandemic is highlighting the weaknesses and deficiencies of the world's food systems [27]. Globally, the issue of food insecurity has emerged as a serious challenge for policymakers [37,43]. The primary difficulty facing the globe today is ensuring that millions of households living in poverty have access to enough food to maintain a healthy life [11]. Policymakers in developing nations, where many people suffer from micronutrient deficiencies and extreme poverty, face a significant challenge in increasing food production and reducing food insecurity despite population growth and climate change [2]. Although it affects

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https://doi.org/10.1016/j.sciaf.2022.e01485
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many different countries and locations, food insecurity is a global phenomenon [21,35]. Due to growing population and rising levels of poverty sub-Saharan African (SSA), food insecurity has gotten worse over time [14]. The bulk of the population lives in rural regions and depends on small-scale rain-fed agriculture for a living, which exacerbates food security challenges in SSA ([20]; Adetoro et al., 2022b).

The agricultural sector in SSA is significantly dominated by smallholder farmers, but their ability to produce is limited by lack of access to cash, productive resources, and suitable technologies to increase food production in the area (Motsoari et al., 2015). One of the SSA countries with the most advanced economies is South Africa. However, there is still a problem with food insecurity in the nation [3,23]. The government of South Africa faces a significant challenge with regard to household food security (Mkhawan et al., 2016). Some of the factors causing food insecurity in South Africa include a lack of employment opportunities (especially in rural areas), rising living expenses, a lack of investment in agricultural development, an increase in informal settlements, and high dependency ratios, particularly in low-income households (Altman and Ngandu, 2010; Ngema et al., 2018; [6]). Agriculture, which plays a significant role in improving South African food security accounts for about 4% of the country's GDP [28].

Most food-insecure households live in poverty, have few or no assets, have no land or just a very small plot of land, and often have a high dependency ratio. Moreover, they are highly vulnerable to external shocks: when they experience such shocks, they often have to adopt "negative" coping strategies (FAO, 2015). Raising the living standards of these households is likely to make the biggest contribution to alleviating hunger and, broadly speaking, food insecurity. This opens ample scope for social protection, that is, "policies and actions which enhance the capacity of poor and vulnerable people to escape from poverty and enable them to better manage risks and shocks" (OECD, 2009). It can have both a "preventional function," that is, preventing households just above the poverty line from falling into the poverty trap by helping them to better manage risks and deal better with shocks, and a "protective function," that is, lifting households above the poverty line.

Using cash transfers as a policy tool, households can become more resilient in their ability to receive food [4]. Cash transfers allow recipient households access to money to enhance their level of living and household welfare because of their latitude and flexibility in use [16,24]. Understanding the effects of financial transfers on enhancing household food security in developing nations, particularly in relation to enhancing their agricultural activities, is a topic of growing discussion and interest (Davis et al., 2016). The main goals of cash transfer programs are to increase agricultural production, reduce poverty, and increase food consumption in rural communities (Alinovi et al., 2009).

Access to productive resources can be difficult for rural households, who depend on agriculture for a living, and this can have a significant impact on both household welfare and agricultural output. Programs for cash transfers can help lower-income households' access to food by increasing agricultural productivity [5,10]. In order to combat hunger and poverty in less developed nations, cash transfer programs are essential (Burchi et al., 2016). Resilient community participation has been stressed in previous studies, with a focus on the socially and economically disadvantaged population, such as the elderly, children, households without money, and the disabled (Bonilla et al., 2017; Tonguet-Papucci et al., 2017).

Empirics show that most cash transfer funders were still unsure of how cash transfer initiatives affected household food security in Africa (see [12,19,26,39]). Additionally, discussions of cash transfer programs in South Africa sometimes focus only on the technical features of the program, design response techniques, and how the money was spent [36,41]. Similar to how these conversations frequently ignore the effects on household food security in favor of the distribution method and cost-effectiveness.

The impact of cash transfer programs on household food security in South Africa is thus not fully recognized. Cash transfers improves the purchasing power of low-income households in South Africa to enhance their standard of living, but they may not always address the underlying reasons of their poverty and food insecurity [7]. It is yet unknown whether cash transfers help low-income families improve their household food security in areas with a high rate of poverty and a lack of formal employment. Studies on the impact of cash transfers programs on household food security have been conducted throughout Africa, with various degrees of success (e.g. [4,9,24,38,40]). Cash transfer schemes in four SSA nations were evaluated by Tiwari et al. [40], who found a range of outcomes.

In addition, d'Agostino et al. [7] examined the impact of cash transfers on household food security using a fuzzy regression discontinuity approach. According to the study, the beneficiaries' household food security was not materially improved by the program. There is a dearth of knowledge on effect of cash transfers on household food security in South Africa while accounting for both endogeneity and selectivity biases in the analysis, as significant as these findings may be. In order to close this gap, this study estimates factors influencing household food security in South Africa using an endogenous poisson model while accounting for biases related to endogeneity and selectivity.

Conceptual framework

This section presents the concepts of household food security measurement and the theoretical model on which the study is hinged on.

Food insecurity indicators

The household food insecurity access scale (HFIAS) was used to evaluate the farming household's food security data. The Food and Nutrition Technical Assistance (FANTA) Project of the United States Agency for International Development (USAID)

developed the HFIAS as a measure of food insecurity [23]. The HFIAS evaluates the level of food insecurity (access) over the previous 30 days (Coates et al., 2007). The coded frequency for each of the nine occurrence questions related to household-level food access was added to determine the HFIAS score for each household. The maximum score for each of the nine questions is 3, while the total score, when added together, has a maximum of 27, and a minimum score of 0. A HFIAS score indicates how secure its access to food is, with a higher score indicating greater food insecurity and a lower score indicating greater food security (Coates et al., 2007). The study of Maziya et al. [23] and Ndlovu et al. [30], who employed the tool in their studies in South Africa, served as the inspiration for the selection of the HFIAS instrument. In addition, this study categorised the households into four categories of household food insecurity using the Household Food Insecurity Access Prevalence (HFIAP). Food security, light food insecurity, moderate food insecurity, and severe food insecurity were the four categories.

Theoretical model

The methodological approach to the study draws inspiration from Terza (1998); Miranda (2004) and Donkoh (2020). This study of cash transfer is hinged on the theory of diffusion of innovation. Following Terza (1998), given thei_{th}farmer from a random sample. Consider the i_{th} individual from a random sample$H = \{1..., n\}$. Dependent on a set of explanatory factors x_i, an endogenous dummyG_i, and a random termβ_i, the dependent variableFS_i—which is a count—is supposed to follow a standard poisson distribution

$$f(FS_i/\mathrm{Y}_i) = \frac{\exp\{-\exp(x_i'\beta + \alpha G_i + \mathrm{Y}_i)\}\{\exp(x_i'\beta + \alpha G_i + \mathrm{Y}_i)\}^{FS_i}}{FS_i!} \tag{1}$$

The random term Y_i is commonly interpreted as a variable that summarizes omitted and unobserved variables. In some contexts, Y_i can be also interpreted as a measurement error. Given a vector of explanatory variables z_i (which may contain some or all elements ofx_i), G_i is characterized by an index process

$$G_i = 1 \quad \text{if} \quad z_i'\theta + \nu_i > 0 \,\text{and0ifotherwise} \tag{2}$$

Where θ is a vector of coefficients to be estimated. $w_i$represents all independent variables, and assume$\mathrm{Y}_i$and$\nu_i$are jointly normal with mean zero and covariance matrix

$$\sum = \begin{pmatrix} \sigma^2 & \sigma\rho \\ \sigma\rho & 1 \end{pmatrix}$$

Conditional on $\mathrm{Y}_i G_i$and $FS_i$are independent. Therefore, the joint conditional likelihood density function of FS_iandG_i, with w_i, is expressed in Eq. (3) as:

$$f(FS_{i,}G_i|w_i) = \int_{-\infty}^{\infty} \begin{Bmatrix} G_i f(FS_i|C=1, w_i, \mathrm{Y}_i)\Pr(G_i = 1|w_i, \mathrm{Y}_i \\ +(1-G_i) f(FS_{i,}G=0, w_i, \mathrm{Y}_i)\Pr(G_i = 0|w_i, \mathrm{Y}_i) \end{Bmatrix} f(\mathrm{Y}_i) d\mathrm{Y}_i \tag{3}$$

Empirical model

According to the theoretical Eq. (1), there are two equations that need to be estimated: the equations for household food security and cash transfers, and as a result, there are two dependent variables. These objectives of the study include estimating the factors that affect access to cash transfers and analyzing how cash transfers affect household food security. In a recursive architecture, the household food security equation may be referred to as the substantive equation and the first equation as the selection equation because it is expected that cash transfer drive will affect it (i.e. cash transfer influences household food security but not the other way round). Therefore, given Eq. (1), the dependent variable of this study is the HFIAS score for household food security. Since it is a count variable, the poisson distribution applies. Cash transfers, as well as some characteristics of farms and farmers vis-à-vis institutional factors, are thought to influence household food security. The true impact of and other variables on may not be able to be measured if we estimate the equations for the two variables separately since there may be some unobserved variables that determine both. For instance, a farmer may be able to access cash transfer if they have access to information.

Similarly, this trait might have a favorable impact on the level of food security in his or her household. In this situation, it becomes impossible to distinguish between the impact of the farmer's access to knowledge and the impact of the cash transfer on his or her level of food security. In other words, how can we tell whether the farmers' access to cash transfers is to blame if, by chance, we discover after estimating the two equations separately that cash transfers have had a major influence on food security? Terza's (1998) approach provides a fix for selectivity bias issues in various simultaneous equation models, similar to Heckman's (1979) treatment effect model. It should be understood, too, that the relationship between the availability of cash transfers and one's level of food security does not necessarily imply endogeneity.

The test that determines whether the selection variable (cash transfer in this study) is endogenous or exogenous is well-illustrated by Miranda (2004). Miranda (2004) proposed estimating packages that are comparable but not identical in either case. Endogenous switching poisson was used in this study to analyze the factors that influence cash transfers and how

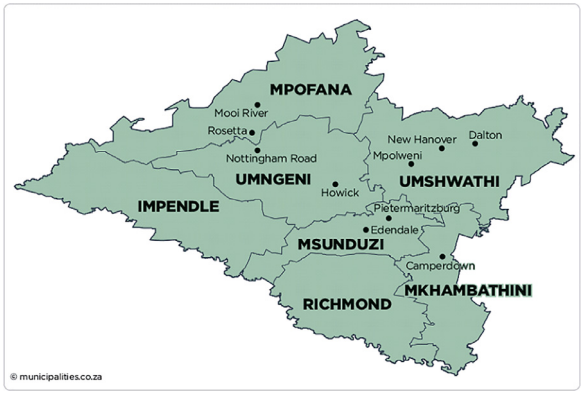


Fig. 1. Map showing the location of the study area at Richmond local municipality.

they affect household food security in the study area. Because it corrects for selectivity bias and endogeneity issues that may develop as a result of various observed and unobserved factors that may affect both cash transfers and household food security, the endogenous switching poisson model proved suitable (Donkoh, 2020). Because the sample may be subject to selection bias and because OLS (Ordinary Least Squares) estimation will produce biased results, two equations were computed to correct the influence of bias in the data (Lin et al., 2019). The access to the cash transfer variable was assumed to be endogenous in the first equation whereas it was assumed to be exogenous in the second.

Study area

Description of the study area

The study was carried out in the Richmond region, namely in the communities of Nkumane, St. Bernard's Mission, Nhlazuka, and Mpofana. The region is situated in the KwaZulu-Natal (KZN) province's midlands, which, along with the Eastern Cape and Limpopo provinces, are thought to be South Africa's three poorest regions [29]. High poverty rates and a lack of possibilities, especially in rural regions, make unemployment rates, which mostly affect young people and account for 42% of all unemployment, even worse (Stats SA, 2012b). Agriculture accounts for 40–45% of the economy, followed by community service (25–39%), trade, and banking (5–10% each). One of the seven local municipalities that make up the larger UMgungundlovu District Municipality is the Richmond Local Municipality, which covers the Richmond area. It is situated about 38 kms south of Pietermaritzburg, KwaZulu-capital Natal's city (IDP, 2015/16).

Understanding the climate in this area is crucial since, among other things, it has a significant impact on the potential for agricultural production. The average temperature in the Richmond area ranges from 14.2 to 16.3 °C. The Mkomazi River Valley has a higher temperature with a range of 17.9–19.8 °C. With humidity, the average maximum temperature of 27 °C in January drops to a minimum of 3 °C in July (Richmond IDP, 2015/2016). In October, a summer rainfall pattern starts, and it lasts until April. About 852 mm of rain may fall on Richmond each year, usually in the middle of the summer. Smallholder farmers typically rely on the rain to irrigate their farmed grounds and frequently have few or no other sources or methods available to them.

The Richmond Local Municipality is divided into 7 wards, with ward 5 housing (Fig. 1) the settlements of Nkumane, St. Bernard Mission, Nhlazuka, and Mpofana. The municipality's population is currently estimated at 65 793 people, who live in 16 446 houses and are growing by 0.4% annually. The majority of the population lives in primarily rural areas that are plagued by a dearth of essential amenities and services as well as by high unemployment. Looking at the level of education, 22% of the population has completed grade 12 (matriculation), while just 4.2% has a higher education. 16.1% of the population has no formal education. In Richmond, households with a female head make up about half of all households (Stats SA, 2017b). According to the Skills Development Framework Review (2009), the majority of Richmond people are employed by businesses with locations in both Pietermaritzburg and Richmond. The findings also state that the neighborhood has a relatively high percentage of low-income earners and that the bulk of the population is heavily reliant on social assistance from the government.

Sampling and data collection

The households based in agriculture who may or may not receive social cash transfers were included in the target population. The target population included agriculture-based households that may or may not be beneficiaries of Social Cash Transfers (SCT). A multistage sampling technique was employed. In the first stage, purposive random sampling method was employed in selecting research participants from a population of households in Richmond Local Municipality. The was based on the ongoing uMngeni Resilience Project (URP) on climate resilience support project which is facilitated by the University

Table 1
Definitions and summary statistics of variables used in the model .

Variables	Description of Variables	Mean	Std. Dev.
Dependent variables Access to Cash Transfer	1 = Access to CT, 0 = otherwise	0.59	
Household food insecurity Explanatory variables	1= food insecure, 0= food secure	0.83	
Age of Household head	Age of HH head (years)	53.33	14.23
Gender Household head	1 if HH head is male, 0 if female	0.63	
Education of Household Head	Years of education of HH head	7.16	3.33
Off-farm income	1 if HH is involved in any off-farm activity	0.54	
Household Size	Number of household size	4.53	2.76
Marital status	1 = if HH married, 0 if otherwise	0.41	
Access to Cooperative	1 if HH has access to Cooperative, 0 if otherwise	0.55	
Access to FBO	1 if HH has access to FBO, 0 if otherwise	0.61	
Farming experience of HH	Years of household experience in farming	6.43	3.84
Access to irrigation facility	1 if HH has access to irrigation, 0 if otherwise	0.37	
Agricultural training Access	1 if HH has access to agricultural trainings, 0 if otherwise	0.36	
Access to Credit	1 if HH has access to credit, 0 if otherwise	0.65	

of KwaZulu-Natal team in Nhlazuka. Due to heterogeneity of the households in the control areas (non- irrigation), systematic random sampling was followed. In the second stage, a simple random sampling was used to select both beneficiaries and non-beneficiaries of SCT in the study with the assistance of the local key informants from the Richmond area. A total sample of 108 households (both beneficiaries and non-beneficiaries) participated in the study.

The study used a quantitative research methodology because its main goal was to determine how different factors related to one another. In this approach, analysis is quantitatively quantified and closed-ended questions are used to convert data into numbers that can be analyzed using statistical techniques. Due to the fact that all respondents were asked the same questions, this enables comparisons of their responses (Crossman, 2014). The use of a quantitative approach in this study makes it possible to pinpoint the impact social cash transfers have on rural households' agricultural output and level of food security. Contrary to the qualitative research approach, a qualitative method frequently maximizes the resources and time used while collecting and analyzing data. This is because it limits the possibility of getting in-depth qualitative analysis of data.

Results and discussion

The descriptive statistics of the households in the study area

This section presents the results and findings of the study. Table 1 presents the description of explanatory variables.

To account for household variability, the socioeconomic parameters of gender, age, educational level, household size, and years of crop farming were incorporated in the model. It has been proposed that these factors may have an impact on the level of food insecurity. About 63% of the 108 respondents were male, while 37% were female. The 53-year-old average age of the farmers in the sample suggests that the majority of the households were in the prime of their lives. The majority of respondents had completed at least their first year of school.

The distribution of household food security status of the farmers

The household food insecurity access prevalence (HFIAP) used in the HFIAS methodology is depicted in Fig. 2. Fig. 2 displays the results, showing that 16.67% of the farmers in the sample were food secure, 40.74% had mild food insecurity, 33.33% had moderate food insecurity, and 9.26% had severe food insecurity. The average HFIAS score of the households was 7.21, indicating that most of the examined farmers experienced food security. A high score suggests that a household is food insecure (De cock et al., 2013).

Impact of access to cash transfer on household food security

For policymakers worldwide, maintaining livelihoods for everyone while ensuring food security has taken precedence. Table 2 shows the findings of the effect of cash transfers on household food security. The estimation findings demonstrate a substantial rho and sigma at 1%, suggesting a correlation between household food security and access to cash transfers. The over-dispersed nature of the data also supports the adoption of the endogenous switching poisson model. It is possible to hypothesize that the endogenous variables can explain the relationship between the variables affecting household food security and access to cash transfers. For brevity, the findings using full information maximum likelihood (FIML) of the endogenous switching poisson model arediscussed.

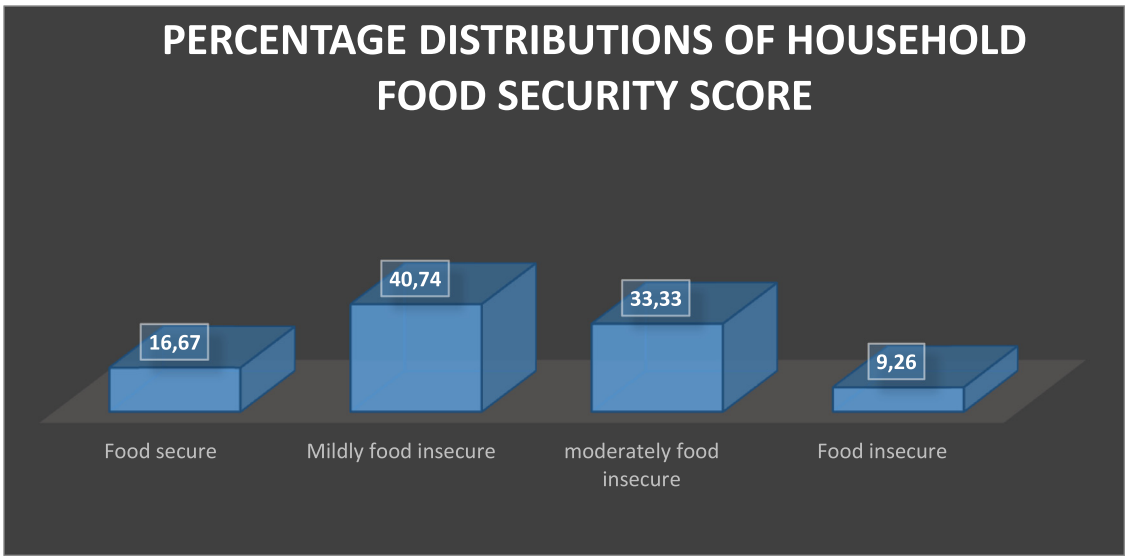


Fig. 2. Percentage distributions of the HFIAS score.

Table 2
FIML estimates of the endogenous switching poisson model for the determinants of access to cash transfer and its impact on household food security.

Household food insecurity status	ENDOGENOUS SWITCHING			EXOGENOUS SWITCHING POISSON		
	Coef.	S. Error	P-value	Coef.	S.Error	P-value
Access to cash transfer	-0.545	0.173	0.002***	-0.438	0.258	0.089*
Gender of Household head	-0.349	0.174	0.045**	0.139	0.244	0.569
Age of Household head	0.046	0.006	0.000***	0.036	0.007	0.000***
Household Size	0.090	0.019	0.000***	0.072	0.023	0.002***
Education Household Head	0.249	0.151	0.100	0.261	0.193	0.176
Access to Credit	-0.404	0.183	0.028**	0.509	−0.231	0.027**
Access to FBO	-0.654	0.322	0.042**	0.519	0.354	0.143
Marital status	0.159	0.072	0.027**	0.103	0.078	0.188
Access to Cooperative	-0.920	0.277	0.001***	0.725	−0.266	0.006***
Agricultural training Access	-0.562	0.179	0.002***	0.498	−0.190	0.009***
Constant	6.535	1.531	0.000***	-5.123	1.467	0.000***
Switch (Cash transfer)						
Age of Household head	0.055	0.013	0.000***	0.050	0.027	0.064*
Access to Credit	-0.369	0.308	0.231	−1.773	0.854	0.038**
Education Household Head	0.111	0.279	0.691	0.836	0.780	0.284
Off-farm income	-1.835	0.964	0.057*	-1.809	1.051	0.085*
Household Size	0.164	0.088	0.062*	0.087	0.112	0.437
Marital status	-0.288	0.188	0.127	-0.134	0.392	0.733
Access to Cooperative	6.613	1236.86	0.996	4.347	226.10	0.985
Access to FBO	0.621	0.714	0.385	-0.414	1.274	0.746
Farming Experience of HH	0.005	0.014	0.737	-0.024	0.026	0.357
Access to irrigation facility	0.085	0.260	0.744	-0.072	0.812	0.930
Constant	-16.29	2473.72	0.995	-6.048	452.21	0.989
Sigma	0.955	0.111	0.000***	0.781	0.099	0.000***
Rho	0.973	0.033	0.000***			
Log likelihood	-161.7			-165.3		
Wald chi2(10)	71.09			1.87		
Prob > chi2	0.0000			0.0000		

*** represent significance level at 1%, 5% and 10%, respectively. FBO means Farm-Based Organization
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* represent significance level at 1%, 5% and 10%, respectively. FBO means Farm-Based Organization.

Findings of Table 2 indicate that all factors, with the exception of the household head's education, are significant at the 1% and 5% levels of significance. The results of endogenous model indicate that access to cash transfers is negative and significant at the 1% level of significance. The availability of cash transfers will improve household welfare and decrease food insecurity; therefore, this outcome was anticipated. Previous research has posited a link between the availability of cash transfers and food security in Africa. Cash transfers encourage investments in agriculture, human capital, education, health, and nutrition, which not only help those in poverty now but also prevent it from happening in the future [13,15,18,25].

The fact that South Africa has not established any cash transfer programs specifically aimed at reducing food insecurity should be highlighted [7]. Instead, they have focused on reducing extreme poverty, which is a major factor in reducing food insecurity. In South Africa, cash transfer programs have successfully reduced hunger and poverty [42]. Cash transfers improve food security by giving low-income households more purchasing power, which boosts their ability to buy food and obtain access to more high-quality food ([1]; Burchi et al., 2016; [5]). This result is in consonance with study of Burchi et al. [5] whose their study showed a positive impact on household dietary variety in Ethiopia, Malawi, and Kenya.

Additionally, the data indicate that the gender of the household head is negatively significant at the 5% level of significance. This suggests that when a man is the leader of the household as opposed to a woman, food insecurity decreases. This finding is in line with that of Bhalla et al. [4], who found that households headed by women typically consumed less food per person than households headed by men. In many houses in developing nations, men are regarded as the "bread winners," and as such, they are expected to provide for the necessities of the family [4]. Male-headed households' food security status will improve as their incomes rise [8,33].

Furthermore, the household head's age has a positive correlation, which means that families with older members are more likely to experience food insecurity than those with younger heads. This finding may be explained by the fact that younger people, as compared to older people, are more active and energetic and can readily seek out off-farm jobs to increase the household's standard of living. This outcome is consistent with studies of Donkoh (2019) and Adetoro et al. (2022a), which found that older farmers were more likely than younger farmers to embrace sustainable agricultural practices, boosting their productivity and household food security.

The household size coefficient is also positive and significant at the 1% level of significance. According to this, larger households are more likely than smaller ones to experience food insecurity. This outcome was anticipated because a larger household is more difficult to sustain than a smaller one, particularly for smallholder farmers who depend on agriculture for their living. This result is in line with that of Awotide et al. (2015), who discovered that the productivity of cassava cultivation in Ghana is negatively impacted by large household sizes. Large household size raises the level of poverty among rural farming households in Nigeria (Omonona and Okunmadewa, 2009).

At a 5% level of significance, the access to credit coefficient is significant and negative. This suggests that farmers who have access to financing are more likely to have a secure supply of food than farmers who do not. Credit-enabled farmers can expand their operations and purchase agricultural inputs, boosting their output and ensuring greater food security. This outcome was anticipated because smallholder farmers typically do not have access to formal financing, which is one of the factors contributing to their food insecurity (Lin et al., 2019; Ojo and Baiyegunhi, [32]; Ojo and Baiyegunhi [34]; Thinda et al., 2021b).

Access to FBO is similarly negative and significant at the 5% level of significance as access to credit. This suggests that compared to farmers who do not belong to an FBO, farmers who belong to a farm base organization are more likely to be in a position of food security. The majority of the time, FBOs offer their members financial support, agricultural inputs, marketing education, and training on farm management. These are crucial for raising farmers' production and ensuring the food security of their households. The findings of Donkoh et al. (2016), Lemessa and Gemechu (2016), Ojo and Baiyegunhi, and (2020a). Access to cooperative and agricultural training is negative and statistically significant at the 1% level. This means that farmers who obtained agricultural training and participated to farmers' cooperative organizations were more likely to have reliable access to food.

The factors that affect cash transfer are shown in the second section of the findings. The findings reveal the importance of household size, off-farm income, and household head's age. Age and household size, however, had favorable and significant effects of 1% and 10%, respectively. The result of this finding is that, compared to younger household heads, older household heads had a larger probability of getting monetary transfers from donors. This finding suggests that the cash transfer is either a "pension" or "old age grant" given to senior citizens in the town. Similar to smaller household sizes, larger household sizes had a higher likelihood of getting cash transfers.

Larger households are more likely to live in poverty, which makes them dependent on social welfare programs like cash transfers to improve their standard of living. Off-farm income, however, was negative and statistically significant at the 10% level. This suggests that households with non-farm incomes are in a better financial position than those whose primary source of income is farming. The outcome is in line with the findings of Ojo and Baiyegunhi [31]; Hlatshwayo et al. [17]. In Nigeria and Kenya, respectively, Babatunde and Qaim [2] and Mathenge and Tschirley [22] observed a positive association between off-farm income and food security and nutrition.

Conclusion and recommendations

An important government policy objective is to strengthen the ability of low-income households in South Africa in order to increase agricultural productivity and household food security. Expanding access to cash transfers will help low-income households gain more influence and eliminate poverty and inequality in the economy. In this study, the effect of access to cash transfers on household food security in rural South Africa was investigated. An endogenous switching poisson model was used to account for both endogeneity and sample selection issues. The findings indicate that factors such as household head gender, access to financing, FBO, cooperative membership, and agricultural training were statistically significant in predicting the level of household food security. In the same vein, the household food insecurity was statistically and significantly influenced by the age of the household head, household size, and marital status. Additionally, the outcome

demonstrates that household size, off-farm income, and household head's age all had a substantial impact on cash transfers. Age and household size had a statistically significant positive impact on household cash transfers, whereas off-farm income had a substantial negative impact. This shows that having access to cash transfers improves the food security of households. As a result, this study recommends that the South African government and other donor organizations to expand cash transfer programs among low-income households in rural South Africa. Sufficiently improved cash transfers, coupled with attention paid to assuring regular, reliable payments, inclusive financial-driven approach appear to be important factors in assuring that cash transfer programmes can alleviate food insecurity and nutrition status. Access to cash transfers is a key component of achieving enhanced household food security.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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